



Web Scraper Overview

A web scraper was developed using **Python Selenium** to collect real-time used car listing data from the Spinny platform across multiple cities.

The scraper automated page navigation, scrolling, and extraction of structured vehicle details such as price, year, fuel type, transmission, kilometers driven, and specifications.

Objective of the Scraper

- Automate the collection of used car listing data from dynamic web pages
- Gather consistent and structured data across multiple cities
- Reduce manual data collection effort
- Build a reliable dataset for market analysis and price prediction modeling

Selenium library imported

```
In [ ]: from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.firefox.options import Options
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
import pandas as pd
import time
```

```
In [ ]: # SETTINGS

MAIN_URL = "https://www.spinny.com/used-cars-in-delhi/s/"
```

```
In [ ]: # SETUP

options = Options()
options.add_argument("--start-maximized")

driver = webdriver.Firefox(options=options)
wait = WebDriverWait(driver, 40)

driver.get(MAIN_URL)
```

```
In [ ]: # CLOSE POPUP

try:
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        popup = WebDriverWait(driver, 8).until(
            EC.element_to_be_clickable((By.XPATH, "//button[contains(., 'later') or
        )
        popup.click()
        print("Popup closed")
    except:
        print("No popup")

```

In []: # LOAD ALL CARS

```

def load_all_cars():

    last_count = 0
    same_count = 0

    while True:

        cards = driver.find_elements(By.CSS_SELECTOR, "div.CarListingDesktop__
        current_count = len(cards)

        print("Cars loaded:", current_count)

        if current_count == last_count:
            same_count += 1
        else:
            same_count = 0

        if same_count >= 3:
            print("All cars loaded.")
            break

        last_count = current_count

        driver.execute_script("window.scrollTo(0, 800);")

        time.sleep(6)

load_all_cars()

cards = driver.find_elements(By.CSS_SELECTOR, "div.CarListingDesktop__carListi
print("Final total cars:", len(cards))

master_data = []

# LOOP THROUGH ALL CARS

for index in range(len(cards)):

    print(f"\nProcessing Car {index+1}")

    try:

```

```

cards = driver.find_elements(By.CSS_SELECTOR, "div.CarListingDesktop__")
card = cards[index]

car_name = card.find_element(By.CSS_SELECTOR, "span.ListingBrandModelD")
car_link = card.find_element(By.TAG_NAME, "a").get_attribute("href")

print("Opening:", car_name)

# Open detail page in new tab
driver.execute_script("window.open(arguments[0]);", car_link)
driver.switch_to.window(driver.window_handles[-1])

time.sleep(3)

# PRICE
price = wait.until(
    EC.visibility_of_element_located(
        (By.XPATH, "//p[contains(@class, 'PriceSectionV3__ogPrice')]")
    )
).text

# OVERVIEW
overview_data = {}
overview_section = wait.until(
    EC.presence_of_element_located((By.XPATH, "//section[@data-categoror
)

overview_items = overview_section.find_elements(
    By.XPATH, ".//div[contains(@class, 'DesktopOverview__overviewItem')]
)

for item in overview_items:
    key = item.find_element(By.XPATH, ".//div[contains(@class, 'itemLab
    value = item.find_element(By.XPATH, ".//div[contains(@class, 'itemD
    overview_data[key] = value

# SPECIFICATIONS
spec_data = {}
spec_section = wait.until(
    EC.presence_of_element_located((By.XPATH, "//section[@data-categoror
)

spec_items = spec_section.find_elements(
    By.XPATH, ".//div[contains(@class, 'styles__carSpecItem')]")
)

for spec in spec_items:
    key = spec.find_element(By.XPATH, ".//p").text
    value = spec.find_element(By.XPATH, ".//h4").text
    spec_data[key] = value

# COMBINE
car_data = {}

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car_data["Car Name"] = car_name
car_data["Price"] = price
car_data.update(overview_data)
car_data.update(spec_data)

master_data.append(car_data)

# Close
driver.close()
driver.switch_to.window(driver.window_handles[0])

time.sleep(2)

except Exception as e:
    print("Error:", e)
    driver.switch_to.window(driver.window_handles[0])
    continue
```

No popup
Cars loaded: 22
Cars loaded: 22
Cars loaded: 22
Cars loaded: 22
All cars loaded.
Final total cars: 22

Processing Car 1
Opening: 2022 Tata Punch

Processing Car 2
Opening: 2020 Maruti Dzire

Processing Car 3
Opening: 2020 Maruti Ciaz

Processing Car 4
Opening: 2019 Audi A6

Processing Car 5
Opening: 2017 Maruti Dzire

Processing Car 6
Opening: 2017 Audi Q3

Processing Car 7
Opening: 2018 Maruti Swift

Processing Car 8
Opening: 2020 Maruti Celerio

Processing Car 9
Opening: 2021 Maruti Wagon R

Processing Car 10
Opening: 2016 BMW 5 Series

Processing Car 11
Opening: 2018 Maruti Swift

Processing Car 12
Opening: 2019 Mercedes E-Class

Processing Car 13
Opening: 2017 Renault Kwid

Processing Car 14
Opening: 2022 Maruti Celerio

Processing Car 15
Opening: 2022 Hyundai Venue N-Line

Processing Car 16

Opening: 2023 Maruti Alto 800

Processing Car 17

Opening: 2015 Hyundai Grand i10

Processing Car 18

Opening: 2023 Maruti Alto 800

Processing Car 19

Opening: 2023 Hyundai Alcazar

Processing Car 20

Opening: 2017 Maruti Alto K10

Processing Car 21

Opening: 2016 Maruti Alto K10

Processing Car 22

Opening: 2020 Kia Seltos

Dataset saved successfully

```
In [ ]: # SAVE DATASET

df = pd.DataFrame(master_data)

df.to_csv("spinny_master_Delhi_dataset.csv", index=False)
df.to_excel("spinny_maste_Delhi_dataset.xlsx", index=False)

print("\nDataset saved successfully")

driver.quit()
```